

CHAPTER 12

Quadrilaterals

NCERT Exercise Solutions

Think and Reflect · Introduction	1 Q
Think and Reflect · 12.1 What Exactly is a Quadrilateral?	3 Q
Exercise Set 12.1	7 Q
In-text questions · 12.2 Parallelograms	3 Q
Exercise Set 12.2	8 Q
In-text questions · 12.3 Applications of Parallelograms	9 Q
Exercise Set 12.3	9 Q
In-text questions · 12.4 Tiling the Plane Using Any 4-gon	4 Q
Exercise Set 12.4	2 Q
End-of-Chapter Exercises	41 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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🤔 Think and Reflect · Introduction

? **Can we use any given quadrilateral to tile the plane? If not, which quadrilaterals can be used and which cannot?**

✔ ANSWER

Yes. Every quadrilateral (4-gon), even a non-convex one, can tile the plane.

- The four angles of a 4-gon add up to 360°, so 4 copies (one of each angle) can fit around a point.
- Method: take a copy and rotate it by 180° about the midpoint of each of its sides; repeat for every new copy (see 12.4).
- For a parallelogram it is easy: a grid of two sets of parallel lines.

🤔 Think and Reflect · 12.1 What Exactly is a Quadrilateral?

1 **Should we call NOPE, SILY, DART, CUTS, OPENS and BENT (Fig. 12.2) quadrilaterals? If not, how will you define a quadrilateral to exclude them?**

✔ ANSWER

- **NOPE:** E, P, O are collinear, so it is really a triangle. Not a quadrilateral.
- **SILY:** S, L, Y are collinear and two sides overlap. Not a quadrilateral.

- **DART:** yes, a quadrilateral (non-convex, it has a dent at D).
- **CUTS:** two sides cross each other (self-intersecting). Usually not called a quadrilateral.
- **OPENS:** it has five points and is not closed. Not a quadrilateral.
- **BENT:** its vertices are not in one plane (non-planar). Not a (planar) quadrilateral.

Definition: four distinct points A, B, C, D in **one plane, no three collinear**, joined by AB, BC, CD, DA, where every point other than A, B, C, D lies on **exactly one** of these four segments.

2 **Can we define a quadrilateral ABCD simply as the points on segments AB, BC, CD and DA (like a triangle)? Will 4 distinct non-collinear points always form a quadrilateral as we visualise it?**

✔ ANSWER

No, not always. We need extra conditions:

- no three vertices collinear (rules out NOPE, SILY),
- all four vertices in one plane (rules out BENT),
- sides meet only at their endpoints (rules out CUTS).

3 **Can you see how to rule out figures like OPENS?**

✔ ANSWER

OPENS has 5 points and its last point S is not joined back to O, so it is not closed.

Our definition uses **exactly four points** and **all four segments AB, BC, CD and DA** (the last side DA closes the figure). OPENS does not fit this, so it is ruled out.

📝 Exercise Set 12.1

1 (i) **Let ABCD be a quadrilateral. List all sides adjacent to side AB. List all sides opposite to AB.**

✔ ANSWER

Adjacent to AB: **BC and DA** (they share an endpoint with AB).

Opposite to AB: **CD**.

1 (ii) **List all angles of ABCD adjacent to ∠A. List all angles opposite to ∠A.**

✔ ANSWER

Adjacent to ∠A: **∠B and ∠D**.

Opposite to ∠A: **∠C**.

1 (iii) **Define a pair of opposite sides and a pair of opposite angles without using the names of the vertices.**

✔ ANSWER

- **Opposite sides:** two sides of the quadrilateral that have **no common endpoint** (they are not adjacent).
- **Opposite angles:** the internal angles at two vertices that are **not joined by a side** (not adjacent vertices).

***2** **Precisely define the internal angle of a quadrilateral at a given vertex. It should work for a non-convex quadrilateral too. (Hint: use the opposite vertex as well.)**

💡 SAMPLE ANSWER

Let the vertex be A, with adjacent vertices B and D, and opposite vertex C.

The two sides AB and AD make the angle ∠BAD (less than 180°) and the reflex angle 360° − ∠BAD.

- If A does **not** lie inside triangle BCD (formed by the other three vertices), the internal angle at A is **∠BAD**.
- If A **lies inside** triangle BCD (A is the dent), the internal angle at A is the **reflex angle 360° − ∠BAD**.

Example: in DART, D lies inside triangle ART, so the internal angle at D is more than 180°.

3 **In a quadrilateral ABCD, suppose AB ∥ DC. Can ABCD be non-convex? What if we instead assume AB = CD? What if we instead assume ∠A = ∠C?**

✔ ANSWER

AB ∥ DC: No, it must be convex.

- If it were non-convex, one vertex, say D, would lie inside triangle ABC.
- A point inside triangle ABC is closer to line AB than C is.
- But DC ∥ AB, so D and C are at the same distance from line AB. Contradiction. (Same for any other vertex.)

AB = CD: Yes, it can be non-convex. Example: A(0, 0), B(4, 0), C(2, 6), D(2, 2). Here AB = CD = 4 and D is inside triangle ABC (dent at D).

∠A = ∠C: Yes, it can be non-convex.
Example: an arrowhead (dart) symmetric about BD: A(−3, 0), B(0, 2), C(3, 0), D(0, 6). By symmetry ∠A = ∠C, and the angle at B is more than 180°.

4 **Take non-collinear points A, B, C and draw lines AB, BC, CA. For every location of D outside these lines, decide if ABCD is self-intersecting, non-convex or convex. (The three lines divide the plane into 7 regions.)**

✔ ANSWER

The 7 regions: inside the triangle; 3 regions across a side; 3 regions at a vertex (opposite the angle of the triangle).

Where D is	ABCD is
Inside triangle ABC	non-convex (dent at D)
Across side AC (opposite B)	**convex**
Across side AB (opposite C)	self-intersecting (CD crosses AB)
Across side BC (opposite A)	self-intersecting (DA crosses BC)
Vertex region at A	non-convex (dent at A)
Vertex region at B	non-convex (dent at B)
Vertex region at C	non-convex (dent at C)

So: **1 region convex, 2 regions self-intersecting, 4 regions non-convex.**

5 **Can a quadrilateral be both self-intersecting and non-planar?**

✔ ANSWER

No. If it is self-intersecting, two opposite sides (say AB and CD) cross at a point.

Two intersecting lines lie in one plane, so A, B, C, D all lie in that plane.

So the quadrilateral is planar.

❓ In-text questions · 12.2 Parallelograms

? **Converses: (a) If the opposite sides of a quadrilateral are equal, it is a parallelogram. (b) If the opposite angles are equal, it is a parallelogram. (c) If the diagonals bisect each other, it is a parallelogram. Are these true?**

✔ ANSWER

All three converses are true.

- (a) Theorem 2: $\Delta ACD \cong \Delta CAB$ by SSS, so alternate angles are equal → AB ∥ DC, AD ∥ BC.
- (b) Theorem 3: $2(\angle A + \angle B) = 360°$, so $\angle A + \angle B = 180°$ → BC ∥ AD; similarly AB ∥ DC.
- (c) Theorem 4: $\Delta AED \cong \Delta CEB$ by SAS → AD ∥ BC; similarly AB ∥ DC.

T&R **If the opposite angles of a quadrilateral are equal, what can we say about adjacent angles? Is the converse true? Fill in: "If each pair of adjacent angles in a quadrilateral ABCD ... then ABCD is a parallelogram."**

✔ ANSWER

$\angle A = \angle C$ and $\angle B = \angle D$, and $\angle A + \angle B + \angle C + \angle D = 360°$.

So $2(\angle A + \angle B) = 360° \rightarrow \angle A + \angle B = 180°$. **Every pair of adjacent angles adds up to 180°.**

Converse: if $\angle A + \angle B = 180°$ and $\angle B + \angle C = 180°$, then $\angle A = \angle C$; similarly $\angle B = \angle D$. **Yes, it is true.**

Filled in: "If each pair of adjacent angles in a quadrilateral ABCD **adds up to 180° (is supplementary)**, then ABCD is a parallelogram."

? Theorem 5 (AB ∥ DC, AB = DC): why must the diagonals AC and BD intersect?

✔ ANSWER

Since AB ∥ DC, ABCD cannot be non-convex (Exercise Set 12.1, Q3), so it is **convex**.

The diagonals of a convex quadrilateral always intersect.

📝 Exercise Set 12.2

1 (i) True or false? A parallelogram with a right angle is a rectangle.

✔ ANSWER

True. Adjacent angles of a parallelogram add up to 180°, so if one angle is 90°, its neighbours are 90° and the opposite angle is 90°. All angles are 90°.

1 (ii) **True or false? A rhombus with perpendicular diagonals is a square.**

✔ ANSWER

False. The diagonals of **every** rhombus are perpendicular, but not every rhombus is a square (its angles need not be 90°).

1 (iii) **True or false? If the diagonals of a parallelogram are equal, then it is a rectangle.**

✔ ANSWER

True. In parallelogram ABCD with AC = BD:

ΔABC ≅ ΔDCB (SSS: AB = DC, BC common, AC = DB).

So ∠B = ∠C. But ∠B + ∠C = 180° (adjacent angles), so ∠B = ∠C = 90°. It is a rectangle.

2 **The diagonal AC of a parallelogram ABCD bisects ∠A. Show that it also bisects ∠C and that ABCD is a rhombus.**

✔ ANSWER

Given ∠DAC = ∠CAB.

AB ∥ DC with transversal AC: ∠CAB = ∠ACD (alternate angles).

AD ∥ BC with transversal AC: ∠DAC = ∠ACB (alternate angles).

So ∠ACD = ∠CAB = ∠DAC = ∠ACB → **AC bisects ∠C**.

In ΔADC, ∠DAC = ∠DCA, so AD = DC (sides opposite equal angles).

Opposite sides of a parallelogram are equal, so AB = DC = AD = BC. **ABCD is a rhombus.**

3 (i) **If the diagonals of a quadrilateral ABCD bisect its angles, must ABCD be a rhombus?**

✔ ANSWER

Yes.

AC bisects ∠A and ∠C: ΔABC ≅ ΔADC (ASA, AC common) → AB = AD, CB = CD.

BD bisects ∠B and ∠D: ΔBAD ≅ ΔBCD (ASA, BD common) → AB = CB, AD = CD.

So AB = BC = CD = DA → rhombus.

3 (ii) **If the diagonals of a quadrilateral ABCD bisect each other at right angles, must ABCD be a rhombus?**

✔ ANSWER

Yes. Diagonals bisect each other → parallelogram (Theorem 4).

Let them meet at E. ΔAEB ≅ ΔAED (SAS: EB = ED, ∠AEB = ∠AED = 90°, AE common) → AB = AD.

A parallelogram with two adjacent sides equal has all sides equal → rhombus.

3 (iii) **If the diagonals of a quadrilateral ABCD are of equal length, must ABCD be a rectangle?**

✔ ANSWER

No. Example: an isosceles trapezium (or a kite) can have equal diagonals but is not a rectangle.

Extra condition: **the diagonals also bisect each other.** Then it is a parallelogram with equal diagonals, so a rectangle (Q1 (iii)).

4 **ABCD is a parallelogram with AB ≠ BC. Show that the pairwise intersection points of the four angle bisectors form the vertices of a rectangle (Fig. 12.11). Why did we assume AB ≠ BC?**

✔ ANSWER

The vertices P, Q, R, S are where bisectors of **adjacent** angles meet (A and B, B and C, C and D, D and A).

$\angle A + \angle B = 180°$, so $\frac{1}{2}\angle A + \frac{1}{2}\angle B = 90°$.

In the triangle formed by A, B and the meeting point of their bisectors, the third angle = 180° − 90° = **90°**.

In the same way, the angle at each of the four meeting points is 90°.

A quadrilateral with all four angles 90° has equal opposite angles → parallelogram (Theorem 3) with a right angle → **rectangle**.

Why AB ≠ BC: if AB = BC, ABCD is a rhombus; its angle bisectors are its diagonals, and all four meet at **one point** (the centre). Then there is no rectangle.

? In-text questions · 12.3 Applications of Parallelograms

? Cut a triangle along the segments joining the midpoints of its sides. Compare the four smaller triangles with each other and with the original. What do you notice?

✔ ANSWER

All four small triangles are congruent to each other, and each is a smaller copy of the original triangle with sides half as long.

? Guess about the segment joining the midpoints of two sides: its direction and length compared with the third side.

✔ ANSWER

It is **parallel to the third side** and **half its length** (Midpoint Theorem).

T&R What can we say about the line that passes through the midpoint of one side of a

triangle and is parallel to another side?

✔ ANSWER

It **bisects the third side** (Theorem 7, converse of the Midpoint Theorem).

(Why?) **In the proof of Theorem 7: why is ΔAPQ ≅ ΔCRQ?**

✔ ANSWER

- AP = BP = CR (BCRP is a parallelogram, so CR = BP).
- ∠AQP = ∠CQR (vertically opposite angles).
- ∠APQ = ∠CRQ (alternate angles, since CR ∥ BA).

So ΔAPQ ≅ ΔCRQ by **AAS**.

? **In Grade 10, P need not be the midpoint of AB; only PQ ∥ BC. Guess the conclusion.**

✔ ANSWER

AP/PB = AQ/QC: the line parallel to BC divides the other two sides in the same ratio (Basic Proportionality Theorem).

? **Draw the three medians of several triangles. What do you see? Compare CM and MP.**

✔ ANSWER

The three medians always pass through **one common point** M, and **CM : MP = 2 : 1** (same for the other medians).

(How?) **Why is ΔMPQ ≅ ΔMYX by ASA (Fig. 12.18)?**

✔ ANSWER

- PQ = XY (both = BC/2) and PQ ∥ XY.
- ∠MPQ = ∠MYX (alternate angles, transversal PY, i.e. line CP).
- ∠MQP = ∠MXY (alternate angles, transversal QX, i.e. line BQ).

So **ASA** (side PQ = YX between the equal angles).

T&R **Another proof of the Centroid Theorem: M is where medians CP and BQ meet. Extend AM to S so that BSCM is a parallelogram. Where should M be on AS? Does this also prove MB ∥ CS?**

✔ ANSWER

Choose S on AM extended so that **M is the midpoint of AS** (MS = AM).

In ΔABS: P is the midpoint of AB, M of AS → PM ∥ BS (Midpoint Theorem). PM lies along MC, so **MC ∥ BS**.

In ΔACS: Q is the midpoint of AC, M of AS → QM ∥ CS. QM lies along MB, so **MB ∥ CS**.

So BSCM is a parallelogram; its diagonals BC and MS bisect each other at X.

Hence X, where line AM meets BC, is the **midpoint of BC**: AX is the third median and passes through M.

Also BS = 2PM and BS = MC, so CM = 2MP (ratio 2 : 1).

? **Join the midpoints of the sides of a quadrilateral. What does the new quadrilateral look like?**

✔ ANSWER

It is always a **parallelogram** (Theorem 9, the Varignon parallelogram).

📝 Exercise Set 12.3

1 (i) **P, Q, R are the midpoints of sides AB, AC, BC of ΔABC. Show that ΔPQR is congruent to ΔQPA and to two other triangles which you should identify.**

✔ ANSWER

By the Midpoint Theorem:

- PQ = BC/2 = BR = RC
- QR = AB/2 = AP = PB
- RP = AC/2 = AQ = QC

ΔPQR and ΔQPA: PQ = QP, QR = PA, RP = AQ → **ΔPQR ≅ ΔQPA** (SSS).

Similarly **ΔPQR ≅ ΔRBP** (PQ = RB, QR = BP, RP = PR) and **ΔPQR ≅ ΔCRQ** (PQ = CR, QR = RQ, RP = QC).

So all four small triangles are congruent.

1 (ii) **Someone erases ΔABC, leaving only ΔPQR. Can you reconstruct ΔABC?**

✔ ANSWER

Yes. Through each vertex of ΔPQR draw a line parallel to the opposite side:

- through P, parallel to QR (this is line AB, since AB ∥ QR),
- through Q, parallel to PR (line AC),
- through R, parallel to PQ (line BC).

These three lines meet at A, B and C.

2 **In ΔABC, M and N are midpoints of AB and AC. D is any point on BC. Show that MN bisects AD.**

✔ ANSWER

By the Midpoint Theorem, MN ∥ BC, so MN ∥ BD.

Let MN meet AD at X.

In ΔABD, M is the midpoint of AB and MX ∥ BD.

By Theorem 7, X is the midpoint of AD. **So MN bisects AD.**

3 **In quadrilateral ABCD, AB ∥ DC and AB ≠ CD. G and H are midpoints of AC and BD. Prove that GH ∥ AB. (Why did we assume AB ≠ CD?)**

✔ ANSWER

Let E be the midpoint of AD.

In ΔADC: E, G are midpoints of AD, AC → EG ∥ DC, so EG ∥ AB.

In ΔDAB: E, H are midpoints of DA, DB → EH ∥ AB.

There is only one line through E parallel to AB, so E, G, H lie on one line, and **GH ∥ AB**.

Why AB ≠ CD: if AB = CD (and AB ∥ DC), ABCD is a parallelogram (Theorem 5); its diagonals bisect each other, so G = H and there is no segment GH.

4 (i) **P, Q, R, S are midpoints of AB, BC, CD, DA of quadrilateral ABCD. Show that PR and QS bisect each other.**

✔ ANSWER

PQRS is a parallelogram (Theorem 9).

PR and QS are its diagonals, and the diagonals of a parallelogram bisect each other.

4 (ii) **Show that if AC = BD, then PR and QS are perpendicular. Is the converse true?**

✔ ANSWER

PQ = AC/2 and QR = BD/2 (Midpoint Theorem). If AC = BD, then PQ = QR.

A parallelogram with adjacent sides equal is a rhombus, and the diagonals of a rhombus are perpendicular. So **PR ⊥ QS**.

Converse is true: if PR ⊥ QS, the parallelogram PQRS has diagonals bisecting each other at right angles, so it is a rhombus (Ex. 12.2, Q3 (ii)). Then PQ = QR → AC/2 = BD/2 → AC = BD.

5 (i) **PQRS is the Varignon parallelogram of ABCD. Copy only PQRS. Show how you will recreate a congruent copy A′B′C′D′ of ABCD from PQRS.**

✔ ANSWER

- Place A′ in the same position relative to P and S as A was: A′P = AP and A′S = AS (on the outer side).
- B′: extend A′P beyond P so that PB′ = PA′ (P is the midpoint of A′B′).
- C′: extend B′Q beyond Q so that QC′ = QB′.
- D′: extend C′R beyond R so that RD′ = RC′.
- Check: S is then the midpoint of D′A′. Join A′B′C′D′.

5 (ii) **Justify why A′B′C′D′ is congruent to ABCD (show S is collinear with A′ and D′, and similarly for P, Q, R).**

✔ ANSWER

$\Delta A'PS \cong \Delta APS$ (SSS), so $\angle A'PS = \angle APS$.

$PB' = PA' = PA = PB$, and $\angle B'PQ = 180° - \angle A'PS - \angle SPQ = 180° - \angle APS - \angle SPQ = \angle BPQ$.

So $\Delta B'PQ \cong \Delta BPQ$ (SAS, PQ common) → $B'Q = BQ$ and the angles at Q match.

In the same way $\Delta C'QR \cong \Delta CQR$ and $\Delta D'RS \cong \Delta DRS$ (e.g. $\Delta SDR \cong \Delta SD'R$).

At S: $\angle A'SP + \angle PSR + \angle RSD' = \angle ASP + \angle PSR + \angle RSD = 180°$ (A, S, D are collinear), so **A′, S, D′ are collinear**, and $A'S = AS = SD = SD'$.

So every side and every angle of $A'B'C'D'$ equals the matching one of ABCD: they are made of the same congruent pieces (PQRS + four corner triangles). **$A'B'C'D' \cong ABCD$.**

5 (iii) **Show that if PQRS is a square, then AC and BD are perpendicular and equal. Prove the converse.**

✔ ANSWER

$PQ \parallel AC$, $PQ = AC/2$; $QR \parallel BD$, $QR = BD/2$.

If PQRS is a square: $PQ = QR$ → **$AC = BD$**; $PQ \perp QR$ → **$AC \perp BD$**.

Converse: if $AC \perp BD$ and $AC = BD$, then $PQ \perp QR$ and $PQ = QR$.

A parallelogram with a right angle and equal adjacent sides is a **square**.

In-text questions · 12.4 Tiling the Plane Using Any 4-gon

T&R **Can we fit 4 copies of SOME around a common point so that each angle is used once? Can we continue to tile the plane?**

✔ ANSWER

Yes. Angles 1 + 2 + 3 + 4 = 360°, so the four angles fill the full turn around a point with no gap or overlap.

Continuing in the right way (rotating copies by 180° about midpoints of sides) gives a tiling.

? **(1) Describe precisely a procedure to draw the tiling of Fig. 12.28. (2) Why does it work?**

💡 SAMPLE ANSWER

(1) Start with one copy. For each side, rotate the copy by 180° about the midpoint of that side to get a neighbour. Repeat for every new copy.

(2) The rotated copy shares the full side (the midpoint maps the side onto itself). Around every vertex the four different angles 1, 2, 3, 4 meet once each, adding to 360°, so there are no gaps or overlaps.

? Method 1: each new copy can be got by rotating any one of its neighbours, and the new copies fit perfectly. Explain why.

💡 SAMPLE ANSWER

Two 180° rotations (about two different midpoints) together give a **slide (translation)**. So every copy is either the original slid, or the original turned by 180° and slid.

Around each vertex, the angles 1, 2, 3, 4 appear once each (sum 360°), and copies meet along equal full sides, so both neighbours give the same new copy and it fits exactly.

? Which of the three possibilities shown in Fig. 12.27 occurs in the tiling?

💡 SAMPLE ANSWER

The arrangement in which, going round the common point, the copies **alternate** between the original position and the 180°-turned position (green, brown, green, brown), and neighbouring copies share a full side of equal length. (Check the picture in your book.)

📝 Exercise Set 12.4

1 Justify why the plane cannot be tiled with a regular pentagon.

✔ ANSWER

Each angle of a regular pentagon = (5 − 2) × 180° ÷ 5 = **108°**.

In a tiling, the angles meeting at a vertex must add up to 360°.

3 × 108° = 324° (gap of 36°); 4 × 108° = 432° (overlap). 360 ÷ 108 is not a whole number.

(If a corner touches the middle of another pentagon's side, the angles there must add to 180°; 180 ÷ 108 is not whole either.)

So regular pentagons cannot tile the plane.

2 **Draw a non-convex 4-gon DART. Show how we can tile the plane with copies of DART. Both methods work. Which do you prefer?**

💡 **SAMPLE ANSWER**

Method 1: rotate DART by 180° about the midpoint of each side to get its neighbours; repeat for every new copy. Around each vertex the angles D, A, R, T (including the reflex angle) add up to 360°.

Method 2: draw the Varignon parallelogram of DART, make a grid of these parallelograms, and place copies of DART on the grid as in Fig. 12.32; the gaps are again copies of DART.

Preference (sample): **Method 1**, because it needs only one simple step (a half-turn about a midpoint) repeated.

End-of-Chapter Exercises

1 **Using a fact about parallelograms, show how to tile the plane using any given triangle.**

✔ ANSWER

Take triangle ABC and a second copy turned by 180° about the midpoint O of side AC; let B go to D.

Now AC and BD bisect each other at O, so ABCD is a **parallelogram** (Theorem 4).

Parallelograms tile the plane (the grid in the introduction). Each parallelogram is two copies of the triangle, so **the triangle tiles the plane**.

2 **Mark the midpoint of the line drawn on the paper (Fig. 12.33); the horizontal lines are equally spaced. Justify.**

💡 SAMPLE ANSWER

The ends of the line lie on two ruled lines that are 4 spaces apart. **The midpoint is where it crosses the middle ruled line** (2 spaces from each end). (Check the picture in your book.)

Justification: let the line be XY and let it cross the middle line at M.

Drop perpendiculars XU onto the middle line and MV onto the line through Y. XU = MV (equal spacing).

∠XUM = ∠MVY = 90°, and ∠XMU = ∠MYV (corresponding angles, parallel lines).

So ΔXUM ≅ ΔMVY (AAS) → **XM = MY**.

3 **In a self-intersecting quadrilateral ABCD, AB and CD intersect at E. Show that ∠A + ∠B + ∠C + ∠D < 360°. Can you construct ABCD with ∠A + ∠B + ∠C + ∠D = 2°?**

✔ ANSWER

The figure is made of two triangles EBC and EAD.

In ΔEBC: ∠B + ∠C = 180° − ∠BEC.

In ΔEAD: ∠A + ∠D = 180° − ∠AED.

∠AED = ∠BEC (vertically opposite), so ∠A + ∠B + ∠C + ∠D = 360° − 2∠BEC **< 360°**.

Yes, for a sum of 2° we need 2∠BEC = 358°, i.e. **∠BEC = 179°**.

Construction: draw segment AB with E on it; through E draw line CD making ∠BEC = ∠AED = 179° (so CD is almost along AB). Join BC and DA.

4 **In parallelogram ABCD, P and Q are on diagonal BD with DP = BQ (Fig. 12.34). Show that APCQ is a parallelogram.**

✔ ANSWER

Let the diagonals AC and BD meet at O. Then OA = OC and OB = OD.

OP = OD − DP and OQ = OB − BQ. Since OD = OB and DP = BQ, **OP = OQ**.

So in APCQ the diagonals AC and PQ bisect each other at O.

By Theorem 4, **APCQ is a parallelogram**.

5 **A right-triangle cutout (right angle at B) is folded so that A touches B (Fig. 12.35). Show that the crease gives the midpoint of AB and also of AC.**

✔ ANSWER

When A falls on B, the crease is the **perpendicular bisector** of AB: it passes through the midpoint of AB and is perpendicular to AB.

$\angle B = 90°$, so BC $\perp$ AB too. Two lines perpendicular to AB are parallel: crease $\parallel$ BC.

So the crease is the line through the midpoint of AB parallel to BC. By Theorem 7 it **bisects AC**: it meets AC at its midpoint.

6 (i) **Medians AP, BQ, CR of ΔABC meet at M. Cut along the medians to get 6 triangles. Show how to assemble them into 3 congruent triangles. Prove you get a triangle. What are its side lengths?**

✔ ANSWER

Turn ΔMPB by 180° about P: B goes to C (PB = PC) and M goes to M′ on line AP extended, with PM′ = PM.

$\angle MPC + \angle M'PC = \angle MPC + \angle MPB = 180°$, so M, P, M′ are on one line. Hence ΔMPC and the moved $\Delta M'PC$ together form **one triangle MCM′**.

Sides: $MC = \frac{2}{3}\,CR$, $CM' = BM = \frac{2}{3}\,BQ$, $MM' = 2MP = AM = \frac{2}{3}\,AP$ (centroid divides medians 2 : 1).

Do the same with the pairs (ΔMQC, ΔMQA) and (ΔMRA, ΔMRB).

Each new triangle has sides **$\frac{2}{3}$ AP, $\frac{2}{3}$ BQ, $\frac{2}{3}$ CR**, so the three are congruent (SSS).

***6 (ii) Repeat the procedure with each of the 3 triangles from (i). Show that each of the 9 triangles has sides AB/3, BC/3 and AC/3.**

✔ ANSWER

Look at T = $\Delta MCM'$ from (i). Its medians are:

- From C to P (midpoint of MM′): CP = **BC/2**.
- From M to the midpoint Z of CM′: M is the midpoint of AM′ (AM = MM′), so in $\Delta AM'C$, MZ = AC/2 (Midpoint Theorem) → **AC/2**.
- From M′ to the midpoint Y of MC: RM = MY (= ⅓ CR) and AM = MM′, so ARM′Y is a parallelogram (diagonals bisect each other) → M′Y = AR = **AB/2**.

By (i), each triangle got from T has sides ⅔ of T's medians: ⅔ × AB/2 = **AB/3**, ⅔ × BC/2 = **BC/3**, ⅔ × AC/2 = **AC/3**.

The same holds for the other two (congruent) triangles, so all 9 triangles have sides AB/3, BC/3, AC/3.

6 (iii) **Cut a triangle and reassemble it into 2 congruent triangles. (Hint: use median AP in ΔABC and then a median of ΔAPB.)**

✔ ANSWER

Cut along median AP (P is the midpoint of BC). Piece 1: ΔAPC with sides AP, PC, AC.

In ΔAPB, let X be the midpoint of AB and cut along median PX.

Turn ΔPXB by 180° about X: B goes to A, P goes to P′ with X the midpoint of PP′. With ΔPXA it forms ΔPAP′.

Its sides: PA; AP′ = BP = PC; PP′ = 2PX = AC (Midpoint Theorem in ΔABC: PX = AC/2).

So ΔPAP′ ≅ ΔAPC (SSS). **Two congruent triangles.**

7 (i) **Trapezium ABCD, AB ∥ DC; E is the midpoint of AD; a line through E meets BC at F. If EF ∥ AB, show F is the midpoint of BC and EF = (AB + CD)/2.**

✔ ANSWER

Join BD; let it meet EF at M.

In ΔDAB: E is the midpoint of AD and EM ∥ AB → M is the midpoint of BD and EM = AB/2 (Theorem 7).

In ΔBDC: M is the midpoint of BD and MF ∥ DC (EF ∥ AB ∥ DC) → **F is the midpoint of BC** and MF = DC/2.

EF = EM + MF = AB/2 + CD/2 = **(AB + CD)/2**.

7 (ii) **If F is the midpoint of BC, show that EF ∥ AB. Which way is simpler?**

✔ ANSWER

Way 1: let M be the midpoint of BD. In ΔDAB, EM ∥ AB. In ΔBDC, MF ∥ DC, so MF ∥ AB.

Only one line through M is parallel to AB, so E, M, F are on one line and **EF ∥ AB**.

Way 2: draw the line through E parallel to AB. By (i) it meets BC at its midpoint, which is F. So this line is EF and EF ∥ AB.

Way 2 is shorter because it uses (i) directly (either way is fine).

8 **The diagonals of parallelogram ABCD meet at O. A line through O meets AB at P and CD at Q. Show that O is the midpoint of PQ.**

✔ ANSWER

In ΔAOP and ΔCOQ:

- OA = OC (diagonals bisect each other)
- ∠AOP = ∠COQ (vertically opposite angles)
- ∠OAP = ∠OCQ (alternate angles, AB ∥ DC, transversal AC)

So ΔAOP ≅ ΔCOQ (ASA) → **OP = OQ**, O is the midpoint of PQ.

(This congruence proof is the simplest.)

***9** **ABCD is a trapezium with parallel sides AD = 3 cm and BC = 5 cm. E and F are midpoints of the non-parallel sides. Find the ratio of the areas of AEFD and EBCF.**

✔ ANSWER

By Ex. 7, EF ∥ AD ∥ BC and EF = (3 + 5)/2 = **4 cm**.

E is the midpoint of AB, so EF is halfway between AD and BC: if the height of ABCD is h, each part has height h/2.

Area AEFD = ½ × (3 + 4) × h/2 = 7h/4.

Area EBCF = ½ × (4 + 5) × h/2 = 9h/4.

Ratio = **7 : 9**.

10 (i) **4 points A, B, C, D, no three collinear. How many different quadrilaterals do they form if self-intersecting or non-convex ones are allowed?**

✔ ANSWER

There are 4! = 24 orders of the letters, and each quadrilateral has 8 names (4 starting points × 2 directions).

24 ÷ 8 = **3 quadrilaterals**: ABCD, ABDC, ACBD.

10 (ii) **How many of these are self-intersecting? How many are convex?**

✔ ANSWER

Case 1: no point lies inside the triangle of the other three. Then **1 is convex and 2 are self-intersecting**.

Case 2: one point lies inside the triangle of the other three. Then **0 are self-intersecting and 0 are convex**: all 3 are non-convex.

11 (i) **PQ ∥ BC with AP = 1, AQ = √5. If PB = 2, find QC.**

✔ ANSWER

Let X be the midpoint of PB (PX = XB = 1) and draw XY ∥ BC meeting AC at Y.

In ΔAXY: P is the midpoint of AX and PQ ∥ XY → Q is the midpoint of AY → QY = AQ = √5.

In trapezium PQCB: X is the midpoint of PB and XY ∥ BC → Y is the midpoint of QC (Ex. 7) → YC = QY = √5.

QC = **2√5**.

***11 (ii)** **If PB = 1/3, find QC.**

✔ ANSWER

Divide AP into 3 equal parts of 1/3. Then AB is made of 4 equal parts of 1/3 (3 in AP, 1 in PB).

Lines through the division points parallel to BC cut AC into 4 equal parts (Theorem 7 and Ex. 7, used again and again).

AQ = 3 parts = √5 → 1 part = √5/3.

QC = 1 part = **√5/3**.

***11 (iii) If PB = 2/3, find QC.**

✔ ANSWER

Divide AP into 3 parts and PB into 2 parts, each 1/3. AB has 5 equal parts.

Parallels to BC through these points cut AC into 5 equal parts.

AQ = 3 parts = √5 → 1 part = √5/3.

QC = 2 parts = **2√5/3**.

***11 (iv) Show that if AP/PB is a rational number, then AP/PB = AQ/QC.**

✔ ANSWER

Let AP/PB = m/n (m, n positive whole numbers). Let s = AP/m; then AP = m·s and PB = n·s.

Divide AB into m + n equal parts of length s and draw lines through the division points parallel to BC.

These lines cut AC into m + n equal parts (first part by Theorem 7, the rest by Ex. 7 (i) in each strip), each of length t.

PQ is one of these lines, so AQ = m·t and QC = n·t.

So AQ/QC = m/n = **AP/PB**.

12 (i) **ABCD is a parallelogram; M, N are midpoints of AB and CD. Show that DM and BN trisect AC.**

✔ ANSWER

MB = AB/2 = DC/2 = DN and MB ∥ DN → MBND is a parallelogram (Theorem 5) → DM ∥ BN.

Let DM and BN meet AC at X and Y.

In ΔABY: M is the midpoint of AB and MX ∥ BY → X is the midpoint of AY: AX = XY.

In ΔCDX: N is the midpoint of CD and NY ∥ DX → Y is the midpoint of CX: XY = YC.

So **AX = XY = YC**: AC is trisected.

12 (ii) **Use (i) to trisect any segment PQ. Find two other ways, one using Exercise 11 and one using the Centroid Theorem.**

✔ ANSWER

Way 1 (parallelogram): take any point B off PQ. Let O be the midpoint of PQ; extend BO to D with OD = OB. PBQD is a parallelogram (diagonals bisect each other). Join D to the midpoint of PB and B to the midpoint of QD: these lines trisect PQ.

Way 2 (Ex. 11): draw a ray from P; mark 3 equal lengths $PX_1 = X_1X_2 = X_2X_3$. Join X_3Q. Through X_1 and X_2 draw lines parallel to X_3Q; they cut PQ into 3 equal parts.

Way 3 (centroid): draw any line through Q and mark B, C on it with QB = QC. Now PQ is a median of ΔPBC. Draw another median, from B to the midpoint of PC; it meets PQ at the centroid G with PG : GQ = 2 : 1, so GQ = PQ/3. The midpoint of PG gives the other trisection point.

13 (i) **Is Theorem 9 true when the quadrilateral is non-convex or self-intersecting? Explain why.**

✔ ANSWER

Yes. The proof uses only the Midpoint Theorem in ΔABC and ΔADC: PQ ∥ AC, PQ = AC/2 and SR ∥ AC, SR = AC/2.

This does not need the quadrilateral to be convex. So PQ and SR are equal and parallel, and PQRS is a parallelogram (Theorem 5).

13 (ii) **In a very special case the Varignon parallelogram becomes a single segment. When?**

✔ ANSWER

PQ ∥ AC and QR ∥ BD. The figure flattens into a segment when PQ and QR lie on one line, that is, **when AC ∥ BD**.

This can happen only for a self-intersecting quadrilateral, e.g. A(0, 0), B(0, 1), C(2, 0), D(2, 1): the midpoints all lie on the line y = ½.

***13 (iii) Does your reasoning apply even when ABCD is non-planar?**

✔ ANSWER

Yes. Triangles ABC and ADC each lie in their own plane, and the Midpoint Theorem works in each: PQ ∥ AC, SR ∥ AC, each equal to AC/2.

Two lines parallel to the same line are parallel, so PQ ∥ SR and PQ = SR. Two parallel lines lie in one plane, so P, Q, R, S are in one plane and PQRS is a parallelogram.

14 P, Q, R, S are on sides AB, BC, CD, DA of ABCD, and PQRS is a parallelogram. Must P, Q, R, S be the midpoints?

✔ ANSWER

No. Example: ABCD a square of side 4, with AP = BQ = CR = DS = 1.

ΔAPS ≅ ΔBQP ≅ ΔCRQ ≅ ΔDSR (SAS: legs 1 and 3, right angle), so PQ = QR = RS = SP.

Opposite sides are equal, so PQRS is a parallelogram (in fact a square), but P, Q, R, S are not midpoints.

15 (i) **Complete this proof of the Midpoint Theorem: extend PQ beyond Q to S. By how much? Why would ΔAPQ ≅ ΔCSQ help?**

✔ ANSWER

Extend PQ so that **QS = PQ**.

ΔAPQ ≅ ΔCSQ (SAS): AQ = CQ, ∠AQP = ∠CQS (vertically opposite), PQ = SQ.

So CS = AP = PB, and ∠QAP = ∠QCS → CS ∥ AB (alternate angles). (This is why the congruence helps: it moves AP to CS.)

CS ∥ PB and CS = PB → PBCS is a parallelogram (Theorem 5).

So PS ∥ BC, i.e. **PQ ∥ BC**, and PS = BC → **PQ = PS/2 = BC/2**.

15 (ii) **Give a similar proof of the converse of the Midpoint Theorem (Theorem 7), extending PQ to a suitable point S.**

✔ ANSWER

P is the midpoint of AB, PQ ∥ BC, Q on AC. Extend PQ to S so that **PS = BC**.

PS ∥ BC and PS = BC → PBCS is a parallelogram (Theorem 5) → CS = PB = AP and CS ∥ AB.

$\Delta APQ \cong \Delta CSQ$ (AAS): $\angle PAQ = \angle SCQ$ (alternate angles), $\angle AQP = \angle CQS$ (vertically opposite), $AP = CS$.

So **AQ = QC** (Q is the midpoint of AC) and PQ = QS = PS/2 = **BC/2**.

***16 Review the properties of a rhombus/rectangle/square from Grade 8. Formulate a converse of each and decide if it is true.**

💡 **SAMPLE ANSWER**

Converse	True?
All 4 sides equal → rhombus	Yes
Diagonals perpendicular → rhombus	No (a kite)
Diagonals perpendicular and bisect each other → rhombus	Yes
Diagonals bisect the angles → rhombus	Yes (Ex. 12.2, Q3 (i))
All 4 angles equal → rectangle	Yes (each 90°)
Diagonals equal → rectangle	No (isosceles trapezium)
Diagonals equal and bisect each other → rectangle	Yes
Diagonals equal and perpendicular → square	No (e.g. a kite)
Diagonals equal, perpendicular and bisect each other → square	Yes

***17** **Show that the sum of angles of a non-planar quadrilateral is always less than 360°. Can you find one with ∠A + ∠B + ∠C + ∠D = 2°?**

💡 **SAMPLE ANSWER**

Draw diagonal AC. In space, an angle at a corner is less than the sum of the other two angles there: ∠DAB < ∠DAC + ∠CAB, and ∠BCD < ∠BCA + ∠ACD (equality only when the four points are in one plane).

So ∠A + ∠B + ∠C + ∠D < (∠DAC + ∠ACD + ∠D) + (∠CAB + ∠BCA + ∠B) = 180° + 180° = **360°**.

Hinge idea: rotate ΔADC about AC. ∠B and ∠D do not change; ∠A and ∠C get smaller as ΔADC folds towards ΔABC.

For 2°: take congruent thin triangles ABC and ADC with ∠B = ∠D = 0.5°. Fold about AC until ∠A + ∠C = 1° (they shrink towards 0° when fully folded). Then the sum = 0.5° + 0.5° + 1° = **2°**. Yes, it is possible.

***18 (i)** **Third tiling method: slide SOME along its diagonal EO until EO matches E′O′. Prove the exact match of SOME with S′O′M′E′ using four parallelograms.**

💡 **SAMPLE ANSWER**

The slide moves every point by the same distance in the same direction (along EO, by length EO).

So for each vertex X of SOME and its new position X′: XX′ ∥ EO and XX′ = EO.

Then EE′S′S, EE′M′M, EE′O′O and SS′M′M are parallelograms (a pair of equal and parallel opposite sides, Theorem 5).

So S′E′ = SE and S′E′ ∥ SE, M′E′ = ME, S′O′ = SO, M′O′ = MO, ... all sides and directions match: the slid cutout lies exactly on S′O′M′E′. (Check the picture in your book.)

***18 (ii)** **Follow the procedure along each diagonal of the starting 4-gon in both directions to get 4 new copies; repeat to get a 3 by 3 grid of 9 copies.**

💡 **SAMPLE ANSWER**

Activity. Slide the starting copy along diagonal EO forwards and backwards, and along diagonal SM forwards and backwards: 4 new copies. Slide these again to fill the 3 × 3 grid.

You will find the 9 copies meet only at vertices, and the gaps between them are 4-gons congruent to SOME (half-turned copies).

***19 (i)** **For positive numbers a ≤ b ≤ c, a triangle with these sides exists exactly when a + b > c. Check by construction.**

✔ **ANSWER**

Draw segment XY = c. Draw a circle of radius a about X and of radius b about Y.

- If a + b > c, the circles meet at a point Z: ΔXYZ has sides c, a, b.
- If a + b ≤ c, the circles do not meet above the line (at most they touch on XY), so no triangle.

***19 (ii)** **A 4-gon has sides 2, 5, 11. Can the fourth side be 100? 10? 1? What are the possible lengths?**

✔ ANSWER

A 4-gon exists when the longest side is less than the sum of the other three.

- 100: 2 + 5 + 11 = 18 < 100 → **No**.
- 10: longest is 11 and 2 + 5 + 10 = 17 > 11 → **Yes**.
- 1: 2 + 5 + 1 = 8 < 11 → **No**.

Possible fourth side x: x < 18 and 11 < 7 + x (x > 4). So **4 < x < 18**.

***19 (iii)** **For given positive a, b, c, d, how will you decide if there is a 4-gon with these side lengths?**

💡 SAMPLE ANSWER

There is such a 4-gon exactly when the longest of the four is less than the sum of the other three.

Why needed: the longest side joins two vertices, and the path along the other three sides between them is longer than a straight segment.

Why enough: imagine a hinged 4-bar linkage; when the longest is shorter than the other three together, it can be opened into a 4-gon (e.g. choose a diagonal that makes two triangles both satisfying the triangle rule).

***20 (i)** **Define a diagonal of an n-gon. How many diagonals does an n-gon have? Make a table for small n.**

✔ ANSWER

A **diagonal** is a segment joining two vertices that are **not adjacent** (not joined by a side).

n	3	4	5	6	7	8
Diagonals	0	2	5	9	14	20

Pattern of increases: +2, +3, +4, +5, +6.

***20 (ii)** **Guess a formula. How many diagonals are added when the number of sides increases by 1? Justify the formula.**

✔ ANSWER

Formula: n(n − 3)/2. Check: n = 8 → 8 × 5 ÷ 2 = 20.

From n to n + 1 sides: the new vertex joins n − 2 old vertices by diagonals, and the old side between its two neighbours becomes a diagonal: **n − 1** new diagonals.

Justification: each vertex joins n − 3 others (not itself, not its 2 neighbours) → n(n − 3) ends; each diagonal has 2 ends, so n(n − 3)/2 diagonals.

***21** **What is the sum of angles of an n-gon? Find the next few values, then a formula, and prove it.**

✔ ANSWER

n	3	4	5	6	7
Sum	180°	360°	540°	720°	900°

Formula: (n − 2) × 180°.

Proof (convex n-gon): draw all diagonals from one vertex. They divide the n-gon into n − 2 triangles, and the angles of these triangles together make up exactly the angles of the n-gon. So the sum = (n − 2) × 180°.

(A non-convex n-gon can also be cut by diagonals lying inside it into n − 2 triangles, giving the same answer.)

***22 (i)** **Write fully and prove: "If (PRLL) and (HALF) are true, then (P MID) and (Q MID) are true."**

✔ ANSWER

Statement: In ΔABC, P is on AB and Q is on AC. **If PQ ∥ BC and PQ = BC/2, then P and Q are the midpoints of AB and AC.**

Proof: extend PQ to R so that PR = BC. PR ∥ BC and PR = BC → PBCR is a parallelogram → CR = PB and CR ∥ AB.

QR = PR − PQ = BC − BC/2 = BC/2 = PQ.

ΔAPQ ≅ ΔCRQ (ASA): ∠APQ = ∠CRQ (alternate angles, AP ∥ CR), PQ = RQ, ∠AQP = ∠CQR (vertically opposite).

So AQ = QC and AP = CR = PB. **P and Q are midpoints.**

***22 (ii)** **Write the converse of the second sentence (keep the first sentence). Write Theorem 7 in terms of the four named conditions.**

✔ ANSWER

"Suppose P is the midpoint of side AB of ΔABC and Q is a point on side AC. **If PQ ∥ BC, then Q is the midpoint of AC.**" This is Theorem 7.

In named conditions: **If (P MID) and (PRLL), then (Q MID) and (HALF).**

***22 (iii)** **Verify that only one pair of assumptions remains: "If (P MID) and (HALF), then (Q MID) and/or (PRLL)?" Write it in words and answer it.**

✔ ANSWER

The 6 pairs: (P MID, Q MID) = Midpoint Theorem; (PRLL, HALF) = part (i); (P MID, PRLL) = Theorem 7; (Q MID, PRLL) = Theorem 7 with P and Q swapped; (Q MID, HALF) is the same as (P MID, HALF) with P and Q swapped. **Only (P MID, HALF) is left.**

In words: "If P is the midpoint of AB, Q is on AC and PQ = BC/2, must Q be the midpoint of AC and PQ ∥ BC?"

Answer: No, not always. A circle of radius BC/2 about P can meet side AC at two points: the midpoint of AC and another point.

Example: A(0, 0), B(3, 2), C(4, 0); P = (1.5, 1). BC = √5, so BC/2 = √1.25. Q = (1, 0) is on AC and PQ = √(0.25 + 1) = √1.25, but Q is not the midpoint (2, 0) and PQ is not parallel to BC.

***23** **Show using reasoning that each of the three tiling methods covers the whole plane with no gaps and no overlaps.**

💡 SAMPLE ANSWER

What to prove: (a) every point of the plane is in some copy (no gaps); (b) two copies never share inside points (no overlaps).

Method 1 (half-turns): two half-turns make a slide, so all copies are the original or its half-turn, slid by whole-number combinations of two fixed slides. Each slide-block (one copy + one half-turned copy) is a hexagon-like shape with opposite sides equal and parallel, and such blocks fit side to side in rows, rows stack without gaps. Around each vertex the angles 1, 2, 3, 4 add to 360°, so there is no overlap.

Method 2 (Varignon grid): the parallelogram grid covers the plane; each grid parallelogram is matched with one copy, and the 4 corner triangles of each copy exactly fill the gap 4-gons, which are themselves copies (Ex. 12.3, Q5).

Method 3 (slides along diagonals): the slides along the two diagonals make a parallelogram grid of copies, and the gaps are half-turned copies, as in Method 1.

***24** **What fraction of the square is shaded? (Fig. 12.43: each vertex is joined to the midpoint of a side; the four lines make a small square in the middle.)**

✔ ANSWER

The four lines cut the square into: the middle square, 4 small right triangles and 4 trapezium-shaped pieces.

Turn each small right triangle by 180° about the midpoint of the side it touches. It fits onto the next trapezium and makes a square of the **same size as the middle square** (the Midpoint Theorem gives the small leg = half the side of the middle square).

So the big square is made of **5 equal squares**, one of which is shaded.

Shaded fraction = **1/5**.

🎉 **Chapter 12 done! Revise the answers once more tomorrow.**